MIL-B-16166B (OS)
1 September 1972
SUPERSEDING
MIL-B-16166A (Wep)
24 May 1960

MILITARY SPECIFICATION

BRONZE, ALUMINUM: FORGINGS, HEAT-TREATED

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the requirements for heat-treated aluminum bronze die forgings and smith or hand forgings. (See 6.1.)

1.2 Classification.

*1.2.1 Composition. Aluminum bronze forgings shall be furnished in the following compositions as specified (see 6.2):

Composition 1 (copper alloy No. 623)
Composition 2 (copper alloy No. 630).

1.2.2 Form. Aluminum bronze forgings shall be furnished in the form of die forgings and smith or hand forgings as specified. If the form is not specified, the manufacturer may use any method of forging.

1.2.3 Condition. Aluminum bronze forgings shall be supplied in the heat-treated condition.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 9530

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SPECIFICATIONS

Federal

PPP-B-585	Box; Wood, Wirebound
PPP-B-601	Box; Wood-Cleated-Plywood
PPP-B-621	Box; Wood, Nailed and Lock-Corner
PPP-C-650	Crates, Wood, Open and Covered

STANDARDS

Federal

FED-STD-151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM E 8	Tension Tests
ASTM E 10	Brinell Hardness Test
ASTM E 18	Rockwell Hardness Tests

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

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Consolidated Classification Committee

Consolidated Freight Classification Rules

(Applications for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1 Material. The material shall be such as to produce forgings in compliance with all requirements of this specification. Forgings shall be produced from a cast ingot or from as-fabricated (rolled, hand forged, or extruded) stock of suitable quality. When the forgings are produced from a cast ingot, the ingot shall be worked sufficiently to produce a thoroughly wrought metallurgical structure.

3.2 Chemical composition.

3.2.1 The chemical composition of the forgings shall conform to the requirements in table I (see 6.1.1).

*Table I

CHEMICAL COMPOSITION
(Percent maximum, unless shown as a range or minimum)

	Copper alloy No. 623	Copper alloy No. 630
Copper and elements with specific limits (% min)	99.5	99.5
Iron	2.0-4.0	2.0-4.0
Tin	0.6	0.20
Zinc	-	0.30
Aluminum	8.5-11.0	9.0-11.0
Manganese	0.50	1.5
Silicon	0.25	0.25
Nickel	1.0	4.0-5.5

3.2.2 The contractor shall furnish an analysis of each melt in the lot showing the percentage of each element designated in table I.

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3.3 Dimensions. The forgings shall conform to the shape and dimensions specified in the contract or order (see 6.2).

3.4 Heat treatment. All forgings shall be given such uniform heat treatment as the manufacturer desires to produce material conforming to the requirements of this specification.

3.5 Mechanical properties. Heat-treated aluminum bronze forgings shall conform to the requirements of table II.

*Table II

MECHANICAL PROPERTIES

Copper alloy No.	Tensile strength, psi (min.)	Yield[1] strength, psi (min.)	Elongation in 2 inches (% min.)	Reduction in area (% min.)	Brinell hardness 300 kg (min.)
623	90,000	45,000	8	8	190
630[2]	105,000	55,000	10	10	202
630[3]	100,000	50,000	10	10	187

[1]Extension under load, 0.005-inch-per-inch extensions.
[2]Small forgings under 250 pounds and section thickness under 5 inches.
[3]Larger forgings over 250 pounds and section thickness over 5 inches.

3.6 Marking. Each forging shall be stamped with the manufacturer's name or identifying symbol, the drawing number, and forging number.

3.7 Workmanship. The forgings shall be uniform in quality, clean, sound, free from foreign material, pipes, segregation, cracks, laps, and other injurious defects. (See 6.3.)

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all forgings of the same cross section made from the same melt and given the same heat treatment.

*4.3 Sampling.

4.3.1 For packaging, packing, and marking. A random sample of shipping containers for the inspection of 4.4.2 shall be selected in accordance with the provisions of MIL-STD-105, inspection level II, AQL 4.0 percent defective.

4.3.2 Sampling for tests.

4.3.2.1 For chemical analysis. One or more samples shall be selected from each lot from a representative cast sample, an ingot, a representative forging, or a tension test specimen of the lot for chemical analysis (4.4.3.1).

4.3.2.1.1 Location. Samples for chemical analysis shall be taken as follows:

(a) Cast samples shall be poured from the same melt as the ingot before the ingot is poured.

(b) Samples from ingots shall be taken from the top and bottom portions of each forging stock and analyzed separately.

(c) Samples from representative forgings shall be taken parallel to the axis of the forging in its thickest cross section at a point midway between the surface and center of the forging, except that forgings less than 1-1/4-inch thick shall be sampled through the entire cross section.

(d) Samples from tension test specimens shall be taken through the entire cross section.

Each sample shall consist of not less than 2 ounces of material.

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4.3.2.2 For mechanical properties. Two tension and hardness test samples shall be cut from representative forgings of each lot or from bars of the same melt, same cross section, and heat treatment as the forgings of each lot.

*4.3.2.2.1 Specimens. Tensile test specimens shall conform to types R1, R2, R3, R4, or R5, method ASTM E 8. The specimen type with the largest diameter that can be machined from the samples shall be used for test.

4.3.2.2.2 Location. For forgings up to 1-1/2 inches in diameter or thickness, the axis of the tensile test specimen shall coincide with the central axis of the forging; for forgings 1-1/2 inches and over in diameter or thickness, the axis shall be midway between the center and surface of the forging. The axis of the tensile test specimen shall be parallel to the major axis of the forging.

*4.4 Inspection.

4.4.1 Visual and dimensional examination. All forgings shall be inspected to determine conformance to the dimensional requirements of 3.3, marking requirements of 3.6, and workmanship requirements of 3.7.

4.4.2 Preservation, packaging, packing, and marking. Forgings selected in accordance with 4.3.1 shall be inspected to determine conformance to the preservation, packaging, packing, and marking requirements of this specification and the contract or order.

4.4.3 Tests.

4.4.3.1 Chemical composition. Unless otherwise specified in the contract or order (see 6.2), samples shall be analyzed by the contractor to determine conformance to the chemical requirements of 3.2. Specimens shall be tested by wet chemical, spectrographic, or other analytical methods. In the event of dispute, analysis shall be by wet chemical methods. If any sample fails to conform to these chemical requirements, the lot represented by the sample shall be subject to rejection. Test procedures for chemical composition shall be as outlined in FED-STD-151.

*4.4.3.2 Tension tests. Tension tests shall be conducted in accordance with method ASTM E 8. The yield strength shall be determined by the extension-under-load methods.

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4.4.3.3 Hardness. Hardness tests shall be made on the fixture ends of the broken tension test specimens and shall be conducted in accordance with methods ASTM E 10 and ASTM E 18.

5. PREPARATION FOR DELIVERY

5.1 Preparation for shipment. Preparation for shipment shall be Level A, B, or C as specified. (See 6.2.) Individual containers shall contain only material of the same lot. Where practicable, each container shall be of the same uniform size and contain the identical number of items of one lot. Containers shall be designed to fit the contents in a compact manner. Forgings shall be adequately blocked, braced, or otherwise secured to prevent their movement within the shipping containers.

5.1.1 Finished or polished forgings. Finished or polished forgings shall be so packed as to afford adequate protection to the finished surfaces. Where practicable, the forgings shall be boxed. Large polished or finished forgings, when boxing is not practicable, shall have finished or polished surfaces protected with batten strips.

5.1.2 Rough forgings. Unless otherwise specified in the contract or order, rough forgings not subject to damage in shipment may be shipped unpacked or bundled.

5.1.3 Forgings having projections. Forgings having projections that may be damaged in handling shall be boxed or crated or the projections adequately protected by wood batten strips.

5.2 Packing.

5.2.1 Level A.

*5.2.1.1 Small forgings. Forgings weighing up to 250 pounds each shall be packed in wirebound wood boxes, cleated-plywood boxes, nailed wood boxes, or wood crates conforming to PPP-B-585 (class 3), PPP-B-601 (overseas type), PPP-B-621 (class 2), or PPP-C-650, as applicable. Finished forgings shall be adequately protected from mechanical damage. The gross weight of shipping container shall not exceed approximately 600 pounds.

5.2.1.2 Other forgings. Large rough forgings weighing more than 250 pounds each require no packing, except that, when specified for convenience in handling due to size and shape, the items shall be secured

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on skids or pallets. Large finished or polished forgings having projections or surfaces that may be damaged shall be packed as specified in 5.1 through 5.1.3. Containers, when required, shall be as specified in 5.2.1.1.

5.2.2 Level B.

*5.2.2.1 Small forgings. Forgings weighing up to 250 pounds each shall be packed in wirebound wood boxes, cleated-plywood boxes, nailed wood boxes, or wood crates conforming to PPP-B-585 (class 1 or 2), PPP-B-601 (domestic type), PPP-B-621 (class 1), or PPP-C-650, as applicable. Finished forgings shall be adequately protected from mechanical damage. The gross weight of shipping containers shall not exceed approximately 600 pounds.

5.2.2.2 Other forgings. Forgings shall be packed as specified in 5.2.1.2 except, shipping containers, when required, shall be in accordance with 5.2.2.1.

5.2.3 Level C. Forgings shall be so prepared and shipping containers so constructed as to insure safe delivery by common or other carrier to the point of delivery at the lowest rate and afford maximum protection from the normal hazards of transportation. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified in the contract or order each bundle, shipping container, and unpacked forging shall be marked with the name of the material, part number as specified on the drawings, alloy and condition of the forgings, this specification number, the gross weight and quantity, the name of the contractor, and the number of the contract or order.

5.3.1 In addition to the marking specified in 5.3 and special markings required in the contract or order, marking of the shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Aluminum bronze forgings covered by this specification are used as breech block guide plates, guide rails, bearing strips, gears, bushings, cams, and other similar applications where strength, corrosion resistance, and good bearing characteristics in contact with steel or monel are required.

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*6.1.1 Compositions 1 and 2. These compositions cover alpha plus beta structure aluminum bronze alloys having good hot working properties. Composition 2 bronze has better hot working properties than composition 1 bronze. Close control of chemical composition is considered essential for meeting mechanical property requirements on heat treating. For Composition 2 bronzes, aluminum contents of 9.0 to 9.5 percent may not be adequate for insuring consistent conformance with mechanical property requirements after heat treatment of forgings with dimensions over 2 inches in diameter or thickness. Control of aluminum content of composition 2 is therefore essential and must be based on forging section size.

*6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Composition and form required (see 1.2)

(c) Dimensions, shape, and quantity

(d) Place of test, if different from 4.4.3.1

(e) Required level of packing (see 5.2)

(f) Special marking, if different from 5.3

(g) Analysis of each melt in the lot for composition furnished (see 3.2.2).

6.3 The contractor may be required to replace forgings in which injurious vendor related defects are revealed by manufacturing operations performed subsequent to acceptance, without expense to the Government.

*6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-B-16166B (OS), Bronze, Aluminum: Forgings, Heat-Treated

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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